University of South Carolina

ECON 594 Applied Econometric Methods and Data Science Basics with AI

Tuesdays and Thursdays, 8:30 to 9:45 a.m.

Fall 2026

1 Instructor

Instructor	Robert Pettis
Office location	TBC
Office hours	TBC
Email	TBC

2 Course description

This course introduces modern causal inference and its integration with machine learning. One half builds the prediction toolkit these methods depend on; the other half develops the causal framework. The two come together in the second half of the semester through heterogeneous treatment effects, meta-learners, double/debiased machine learning, and causal forests. By the end of the term you should be able to estimate not only whether a treatment worked, but for whom and by how much, and to defend those estimates. All work is done in Python.

3 Prerequisite

ECON436 Introductory Econometrics, or an equivalent course in econometrics or statistics. Familiarity with basic probability and regression is assumed. The Python needed for the course is taught from the first week; no prior Python experience is required.

4 Textbooks

4.1 Required (both free online)

- Matheus Facure, *Causal Inference for the Brave and True*. The main text, free online at https://matheusfacure.github.io/python-causality-handbook/landing-page.html.
- Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, and Jonathan Taylor, *An Introduction to Statistical Learning with Applications in Python* (Springer, 2023). For the machine-learning methods. The free PDF at https://www.statlearning.com/ is identical to the print edition, so a printed copy is an option for those who prefer physical books.

4.2 Optional

- Matheus Facure, *Causal Inference in Python* (O'Reilly, 2023). The professionally edited and expanded version of the free main text above; worth buying if you want a polished print reference.
- Scott Cunningham, *Causal Inference: The Mixtape* (Yale, 2021), free at https://mixtape.scunning.com/. A second edition, *Causal Inference: The Remix*, is due August 25, 2026, for those who prefer to wait for the updated version.

- Victor Chernozhukov, Christian Hansen, Nathan Kallus, Martin Spindler, and Vasilis Syrgkanis, *Applied Causal Inference Powered by ML and AI*. The most direct reference for the double/debiased machine-learning part of the course. Free at https://causalml-book.org/.
- Trevor Hastie, Robert Tibshirani, and Jerome Friedman, *The Elements of Statistical Learning* (Springer, 2009). A deeper reference for the machine-learning methods. Free at https://hastie.su.domains/ElemStatLearn/.
- Josh Starmer, *The StatQuest Illustrated Guide to Machine Learning* (StatQuest Publications, 2022). A friendly, highly visual introduction to the machine-learning methods, good for building intuition.
- Josh Starmer, *The StatQuest Illustrated Guide to Neural Networks and AI* (StatQuest Publications, 2024). The same illustrated approach applied to neural networks.

5 Technology requirements

This course is taught in Python. You will also need git (https://git-scm.com/). A setup guide and the course environment are provided in the course repository. Lectures are Jupyter notebooks, and you will run and edit the same notebooks yourself.

If you have questions or problems with your computer or software, or need technical support (including Blackboard support):

- Contact the Division of Information Technology (DoIT) Service Desk at (803) 777-1800,
- Submit an online request through the Self-Service Portal (https://scprod.service-now.com/sp), or
- Visit the Carolina Tech Zone (https://sc.edu/carolina_tech_zone).

If you experience computer issues, computer labs are available at the Thomas Cooper Library and in certain campus buildings.

6 Assignments and assessment

Item	Weight (undergraduate)	Weight (graduate)
Problem sets	20%	10%
Test 1	10%	5%
Prediction competition *or* Test 2	10%	20%
Final exam	15%	15%
Research project proposal	10%	15%
Research project report	25%	35%
Participation	10%	0%

6.1 Problem sets

There will be at least three problem sets, each requiring some programming in Python. These problem sets are to be submitted to your github repository which will be a fork of the course's

repository. No late problem sets are accepted. Only packages used in the course are allowed, unless you have prior permission.

6.2 Tests

Test 1 is an in-class midterm, sat by everyone on the date in the schedule.

Test 2 is optional. The prediction competition described below occupies that slot by default, and undergraduates who would rather sit a second closed-book test may do so instead. Either way it is the same share of your grade. See the competition section for how to choose.

6.3 Final exam

There will be a cumulative final exam on Tuesday, December 8, at 9:00 a.m., set by the university final examination schedule for a class meeting Tuesdays and Thursdays at 8:30 a.m. Like the midterm tests, it is closed-book and in class, with no AI and no electronic devices.

6.4 Research project

You will investigate a prediction or causal-inference problem using methods from this course.

- The proposal explains the research question, data, and methodology. It is 1 to 3 pages, excluding references.
- The report adds an introduction, literature review, results, and discussion. It is at least 10 pages, excluding references.

Both must be typed on letter-size paper in 12-point font. PhD students should use LaTeX.

6.5 Prediction competition

In place of Test 2 you will enter a prediction competition, run the way Kaggle runs one. You receive the sale records of a Japanese woodblock print auction house and build the best model you can of the log of the final sale price. A few thousand of those lots come to you with the price removed. You predict those, and I mark the predictions against the prices I held back.

- **What you submit.** A file of predictions for the test rows, the code that produced it, and a short report of at most four pages describing what you tried, what you kept, and why.
- **Extra data is allowed and encouraged.** You may crawl, merge, or hand-collect anything that helps, provided every added variable is listed in the report with its source and the date you obtained it. An unsourced variable is treated as an integrity violation, not a modelling choice.
- **Any package is permitted**, including ones we did not use in class. The restriction in the problem sets does not apply here. You must still be able to explain every step of what you submitted.
- **Your score is not your grade.** Accuracy on the held-out set is part of the mark, not the whole of it. The rubric, distributed with the data, weighs method, validation, the honesty of your reporting, and the report itself.

Undergraduates may sit Test 2 instead. If you prefer a second closed-book test to the competition, you may sit one on the date in the schedule. You must tell me which you are doing **no later than one week before the due date**, and after that deadline the choice is fixed. Graduate students all enter the competition. If you say nothing, you are entered in the competition by default.

7 Use of generative AI

Generative AI tools such as ChatGPT, Claude, and Copilot are permitted in this course, and learning to use them well is part of your training. You will use them in research and in your careers, as I do. That permission carries one firm condition: you are responsible for everything you submit. You must understand it, explain it in your own words, and reproduce it on your own.

Keep this in mind: you cannot truly audit what an AI produces without being able to do the work yourself. If you cannot derive the result or write the code unaided, you cannot tell when the tool is wrong, and it is often confidently wrong. Treat AI as a way to work faster and to learn, never as a substitute for understanding, and verify every output against the course material and primary sources.

- **Research project.** AI is permitted and encouraged as a working tool. Disclose which tools you used and for what, and cite any AI-assisted text or code. You are responsible for the accuracy of everything in the report and must be able to defend it.
- **Problem sets.** You may use AI to help you learn, but you must be able to reproduce and explain every step on your own. As stated above, you may not use packages that were not used in the course without permission.
- **Prediction competition.** AI is permitted, as are any packages and any outside data. Disclose the tools you used and for what, in the same place you list your data sources. You must be able to explain and rerun everything you submit, and I may ask you to walk me through it.
- **Tests and final exam.** Closed-book, in class, with no AI and no electronic devices.

Under the USC Honor Code, using AI where it is not authorized, or submitting work you cannot explain, is an academic integrity violation. If you are unsure whether something is allowed, it is your responsibility to ask first.

8 Grading policy

89.5% to 100%	A
84.5% to 89.4%	B+
79.5% to 84.4%	B
74.5% to 79.4%	C+
69.5% to 74.4%	C
64.5% to 69.4%	D+
59.5% to 64.4%	D
0% to 59.4%	F

9 Schedule

The following is a **tentative** schedule of the course.

#	Day	Date	Topic
1	T	8/18/2026	Introduction to Causal Inference; course setup (git, Python, packages)
2	R	8/20/2026	Randomized experiments and stats review
	M	8/24/2026	*Last day to change or drop without "W"*
3	T	8/25/2026	Randomized experiments and stats review
4	R	8/27/2026	Graphical Models
5	T	9/1/2026	Graphical Models
6	R	9/3/2026	Regression and orthogonalization (Frisch, Waugh, and Lovell)
7	T	9/8/2026	Regression and orthogonalization (Frisch, Waugh, and Lovell)
8	R	9/10/2026	Gradient descent
9	T	9/15/2026	Cross-validation
10	R	9/17/2026	**Test 1**
11	T	9/22/2026	Regularization: ridge and lasso
12	R	9/24/2026	Decision trees
13	T	9/29/2026	Ensembles of trees: bagging, random forests, boosting
14	R	10/1/2026	Neural network fundamentals
15	T	10/6/2026	Backpropagation
16	R	10/8/2026	Logistic regression: likelihood, cross entropy, and the one neuron
17	T	10/13/2026	SoftMax and cross entropy; K nearest neighbors and matching
	R	10/15/2026	*Fall Break (no classes)*
18	T	10/20/2026	Convolutional and recurrent networks (survey; if time)
19	R	10/22/2026	Propensity scores
20	T	10/27/2026	Effect heterogeneity and the CATE
21	R	10/29/2026	Meta-learners
	R	10/29/2026	*Competition: tell me by today if you are sitting the test instead*
	T	11/3/2026	*General Election Day (no classes)*
	W	11/4/2026	*Last day to drop without "WF"*
22	R	11/5/2026	**Prediction competition due**, or **Test 2** in class for undergraduates who chose it
23	T	11/10/2026	Double/Debiased Machine Learning
24	R	11/12/2026	Double/Debiased Machine Learning
25	T	11/17/2026	Causal Forests
26	R	11/19/2026	Causal Forests
	T	11/24/2026	*Thanksgiving Recess (no classes)*
	R	11/26/2026	*Thanksgiving Recess (no classes)*
27	T	12/1/2026	Special topics
28	R	12/3/2026	Special topics; last day of class
	T	12/8/2026	**Final exam**, 9:00 a.m.